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- **Unit 1**: DC Network Analysis, Kirchhoff's Laws, Node/Mesh, Thevenin, Norton, Superposition.
- **Unit 2**: AC Circuits, Phasors, Impedance, Series RLC Resonance, 3-Phase Polyphase Systems.
- **Unit 3**: Magnetic Coupling, Transformers, DC Motors & Generators, AC Induction Machines.
- **Unit 4**: Semiconductor Physics, PN Junction, Diodes, Zener Regulators, Rectifiers, MOSFETs.
- **Unit 5**: Passive & Active Filters, Operational Amplifiers, Integrator, Comparator, ABS.
- **Unit 6**: Microprocessor vs Microcontroller, Arduino ATmega328P, Sensors Interfacing, IoT Architecture.

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CHAPTER 1

Fundamentals of D.C. Circuits

1.1 Introduction and Learning Objectives

Direct Current (DC) circuit theory provides the foundational bedrock upon which all electrical, electronic, communications, and computer engineering disciplines are erected. Whether designing high-speed digital motherboards, solar photovoltaic microgrids, or electric vehicle powertrains, an engineer must possess an intuitive and rigorous mastery of circuit topology, constitutive relations of passive elements, Kirchhoff's conservation laws, systematic matrix formulation methods, and classical network reduction theorems.

Unit Learning Objectives

Upon mastering this chapter, the engineer will be capable of:

1. Applying the passive sign convention to ascertain power generation versus dissipation.
2. Utilizing the constitutive relations of resistance, inductance, and capacitance under DC steady-state.
3. Stating and applying Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) to complex planar networks.
4. Performing Star–Delta ($\mathrm{Y}-\Delta$) impedance conversions to simplify non-series-parallel circuits.
5. Formulating systematic Mesh and Nodal matrix equations, resolving supermeshes and supernodes.
6. Applying the Superposition Theorem, Thevenin's Theorem, Norton's Theorem, and the Maximum Power Transfer Theorem to simplify and optimize linear circuits.

1.2 Circuit Variables and Passive Elements

1.2.1 Current, Voltage, Power, and Energy

Definition: Electric Current and Potential

Electric current $i(t)$ is defined as the time rate of flow of net electric charge across a specified cross-sectional boundary:

$$i(t) = \frac{dq(t)}{dt} \quad [\text{Amperes, A}] \iff q(t) = \int_{t_0}^{t} i(\tau)d\tau + q(t_0) \tag{1.1}$$

where 1 A = 1 Coulomb/second. **Electric potential difference** (voltage) $v(t)$ is the energy expended or work done in moving a unit positive charge between two points:

$$v(t) = \frac{dw}{dq} \quad \text{[Volts, V]} \tag{1.2}$$

where 1 V = 1 Joule/Coulomb.

Definition: Instantaneous Power and Passive Sign Convention

The instantaneous power $p(t)$ delivered to or absorbed by any two-terminal electrical element is the product of the terminal voltage and current:

$$p(t) = v(t) \cdot i(t) = \frac{dw}{dq} \cdot \frac{dq}{dt} = \frac{dw}{dt} \quad \text{[Watts, W]} \tag{1.3}$$

Passive Sign Convention:

- When the reference direction for current enters the positive (+) voltage terminal: $p(t) > 0 \implies$ the element **absorbs** (dissipates or stores) electrical power.
- When the current leaves the positive (+) terminal: $p(t) < 0 \implies$ the element **delivers** (supplies) electrical power to the external network.

Theorem / Law: Tellegen's Theorem / Conservation of Power

In any lumped physical electrical network containing B branches, the algebraic sum of instantaneous powers across all branches at any instant is identically zero:

$$\sum_{k=1}^{B} p_k(t) = \sum_{k=1}^{B} v_k(t) i_k(t) = 0 \iff \sum P_{\text{delivered}} = \sum P_{\text{absorbed}} \tag{1.4}$$

1.2.2 Passive Circuit Elements: Resistor, Inductor, Capacitor

Definition: Linear Resistor and Ohm's Law

A resistor is an electrical component that dissipates electrical energy irreversibly into thermal energy. For a linear, time-invariant resistor, the voltage-current relationship is governed by **Ohm's Law**:

$$v(t) = R \cdot i(t) \iff i(t) = G \cdot v(t) \tag{1.5}$$

where R is the electrical resistance in Ohms (Ω) and $G = 1/R$ is the conductance in Siemens (S). The resistance of a uniform conductor of length L, cross-sectional area A, and material resistivity ρ is:

$$R = \rho \frac{L}{A} = \frac{L}{\sigma A} \tag{1.6}$$

Resistance exhibits temperature dependence characterized by the temperature coefficient α_0:

$$R(T) = R_0 \left[1 + \alpha_0 (T - T_0)\right] \tag{1.7}$$

where pure metals exhibit a positive temperature coefficient ($\alpha > 0$, resistance increases with temperature), while semiconductors and carbon exhibit a negative temperature coefficient ($\alpha < 0$).

Definition: Inductance and Capacitor DC Steady-State Behavior

An **inductor** stores energy in an induced magnetic field according to:

$$v_L(t) = L\frac{di_L(t)}{dt}, \qquad W_L(t) = \frac{1}{2}Li_L(t)^2 \tag{1.8}$$

Under steady DC conditions, currents are invariant with time ($di/dt = 0$). Hence, the voltage across an ideal inductor is zero ($v_L = 0$). An ideal inductor behaves as a **short circuit** under DC steady-state.
A **capacitor** stores energy in an electrostatic field:

$$i_C(t) = C\frac{dv_C(t)}{dt}, \qquad W_C(t) = \frac{1}{2}Cv_C(t)^2 \tag{1.9}$$

Under steady DC conditions, voltages are constant ($dv/dt = 0$). Hence, the current through an ideal capacitor is zero ($i_C = 0$). An ideal capacitor behaves as an **open circuit** under DC steady-state.

1.3 Fundamental Circuit Laws and Reduction Methods

1.3.1 Kirchhoff's Laws

Theorem / Law: Kirchhoff's Current Law (KCL)

The algebraic sum of all currents entering any node (or enclosed boundary) in an electric circuit is zero at every instant of time:

$$\sum_{k=1}^{N} i_k(t) = 0 \iff \sum i_{\text{entering}} = \sum i_{\text{leaving}} \tag{1.10}$$

KCL is the direct circuit-theoretic manifestation of the law of **conservation of electric charge**.

Theorem / Law: Kirchhoff's Voltage Law (KVL)

The algebraic sum of all potential differences (voltages) around any closed loop or path in an electric circuit is zero at every instant of time:

$$\sum_{k=1}^{M} v_k(t) = 0 \iff \sum V_{\text{rises}} = \sum V_{\text{drops}} \tag{1.11}$$

KVL is the direct circuit-theoretic manifestation of the law of **conservation of energy**.

1.3.2 Voltage and Current Division Rules

For two resistors R_1 and R_2 connected in series across total voltage V:

$$V_1 = V\frac{R_1}{R_1+R_2}, \qquad V_2 = V\frac{R_2}{R_1+R_2} \tag{1.12}$$

For two resistors R_1 and R_2 connected in parallel carrying total current I:

$$I_1 = I\frac{R_2}{R_1+R_2}, \qquad I_2 = I\frac{R_1}{R_1+R_2} \tag{1.13}$$

1.3.3 Star–Delta (Y – Δ) Transformations

When three resistors form a closed three-terminal configuration that cannot be decomposed into simple series or parallel combinations, equivalence transformations are mandatory.

Theorem / Law: Delta-to-Star (Δ → Y) Conversion

Given delta-connected resistors R_{AB}, R_{BC}, R_{CA}, the equivalent star branch connected to node A is:

$$R_A = \frac{R_{AB}R_{CA}}{R_{AB}+R_{BC}+R_{CA}}, \quad R_B = \frac{R_{AB}R_{BC}}{R_{AB}+R_{BC}+R_{CA}}, \quad R_C = \frac{R_{BC}R_{CA}}{R_{AB}+R_{BC}+R_{CA}} \tag{1.14}$$

If all delta resistors are identical (R_Δ), then $R_Y = \frac{R_\Delta}{3}$.

Theorem / Law: Star-to-Delta (Y → Δ) Conversion

Given star-connected resistors R_A, R_B, R_C, the equivalent delta resistor across terminals $A-B$ is:

$$R_{AB} = R_A+R_B+\frac{R_AR_B}{R_C}, \quad R_{BC} = R_B+R_C+\frac{R_BR_C}{R_A}, \quad R_{CA} = R_C+R_A+\frac{R_CR_A}{R_B} \tag{1.15}$$

If all star resistors are identical (R_Y), then $R_\Delta = 3R_Y$.

1.4 Systematic Network Analysis: Mesh and Nodal Methods

1.4.1 Mesh Current Analysis

Mesh analysis applies KVL to planar networks where loop boundaries do not intersect:

$$[\mathbf{R}]\mathbf{I}_{\text{mesh}} = \mathbf{V}_{\text{mesh}} \implies \begin{bmatrix} R_{11} & -R_{12} & \cdots \\ -R_{21} & R_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} V_{s1} \\ V_{s2} \\ \vdots \end{bmatrix} \tag{1.16}$$

where R_{ii} is the sum of resistances in mesh i, R_{ij} is the mutual resistance shared by mesh i and j, and V_{si} is the net clockwise voltage rise in mesh i.

Definition: Supermesh

When a current source is situated in a branch shared by two adjacent meshes, a **supermesh** is created by removing the branch containing the current source (opening the branch) and writing a single unified KVL equation around the perimeter of both meshes, complemented by an auxiliary branch current constraint equation:

$$I_2 - I_1 = I_{\text{source}} \tag{1.17}$$

1.4.2 Nodal Voltage Analysis

Nodal analysis designates one reference ground node (0 V) and applies KCL at all $N-1$ non-reference nodes:

$$[\mathbf{G}]\mathbf{V}_{\text{node}} = \mathbf{I}_{\text{node}} \implies \begin{bmatrix} G_{11} & -G_{12} & \cdots \\ -G_{21} & G_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} i_{s1} \\ i_{s2} \\ \vdots \end{bmatrix} \tag{1.18}$$

where G_{ii} is the sum of conductances incident to node i, G_{ij} is the mutual conductance between nodes i and j, and i_{si} is the net current injected into node i.

Definition: Supernode

When an ideal voltage source is connected directly between two non-reference nodes without series resistance, a **supernode** is formed by enclosing the voltage source and both adjacent nodes in a generalized Gaussian boundary, writing a combined KCL equation complemented by the constraint $v_2 - v_1 = V_{\text{source}}$.

1.5 Network Theorems

Theorem / Law: Superposition Theorem

In any linear bilateral electrical network containing two or more independent sources, the voltage across or current through any branch is equal to the algebraic sum of the voltages or currents produced by each independent source acting individually, with all other independent sources deactivated:

- Independent voltage sources are **deactivated by short-circuiting** ($v = 0$).
- Independent current sources are **deactivated by open-circuiting** ($i = 0$).
- Dependent sources must remain fully connected and active.

Theorem / Law: Thevenin's and Norton's Theorems

Any linear, bilateral two-terminal DC network can be replaced across its load terminals $A - B$ by:

1. **Thevenin Equivalent**: An ideal voltage source V_{th} in series with resistance R_{th},

where:

$$V_{th} = V_{oc} \quad \text{(Open-circuit voltage across terminals } A-B\text{)} \tag{1.19}$$

$$R_{th} = \text{Input resistance across } A-B \text{ with all independent sources deactivated} \tag{1.20}$$

2. **Norton Equivalent**: An ideal current source I_N in parallel with resistance R_N, where:

$$I_N = I_{sc} \quad \text{(Short-circuit current flowing from } A \text{ to } B\text{)}, \qquad R_N = R_{th} \tag{1.21}$$

The two representations are related via source transformation:

$$V_{th} = I_N R_N \iff I_N = \frac{V_{th}}{R_{th}} \tag{1.22}$$

Theorem / Law: Maximum Power Transfer Theorem

A variable resistive load R_L connected across a linear two-terminal DC network receives maximum electrical power when the load resistance equals the internal Thevenin resistance of the driving network:

$$R_L = R_{th} \tag{1.23}$$

Under this matched condition, the maximum power transferred is:

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{I_N^2 R_N}{4} \tag{1.24}$$

The transfer efficiency under maximum power is strictly $\eta = \frac{P_{load}}{P_{total}} \times 100\% = 50\%$.

Worked Example: Thevenin Reduction with Independent and Dependent Sources

Determine the Thevenin equivalent circuit across terminals $A-B$ for a network driven by a 20 V independent source, an $8\,\Omega$ resistor, a dependent current source $0.5I_x$ in parallel with a $4\,\Omega$ resistor, where I_x is the current through the $8\,\Omega$ branch.

Solution:

1. **Open-Circuit Voltage** V_{oc}: With terminals $A-B$ open, the current I_x flows from the 20 V source through $8\,\Omega$ and through the parallel combination. KCL at node A:

$$I_x + 0.5I_x = \frac{V_A}{4} \implies 1.5I_x = 0.25V_A$$

Since $I_x = \frac{20-V_A}{8}$, substitute:

$$1.5\left(\frac{20-V_A}{8}\right) = 0.25V_A \implies \frac{30-1.5V_A}{8} = 0.25V_A \implies 30-1.5V_A = 2V_A \implies 3.5V_A = 30$$

$$V_{th} = V_A = \frac{30}{3.5} \approx 8.57 \text{ V}$$

2. **Short-Circuit Current** I_{sc}: Connecting a short between A and B, $V_A = 0$ V. The current through the $4\,\Omega$ resistor is zero. $I_x = \frac{20-0}{8} = 2.5$ A. The dependent source

delivers $0.5I_x = 0.5(2.5) = 1.25$ A. By KCL at the shorted node: $I_{sc} = I_x + 0.5I_x = 1.5(2.5) = 3.75$ A.

3. **Thevenin Resistance** R_{th}:

$$R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{8.5714 \text{ V}}{3.75 \text{ A}} \approx 2.286\,\Omega$$

1.6 Chapter Summary

- Passive elements obey: $v_R = Ri_R$, $v_L = L\frac{di_L}{dt}$ (short at DC), $i_C = C\frac{dv_C}{dt}$ (open at DC).
- KCL states $\sum I = 0$ (charge conservation); KVL states $\sum V = 0$ (energy conservation).
- Star–Delta conversion scales identical resistors by a factor of 3: $R_\Delta = 3R_Y$.
- Thevenin and Norton equivalents provide compact linear driving-point models: $V_{th} = V_{oc}$, $I_N = I_{sc}$, $R_{th} = R_N = V_{oc}/I_{sc}$.
- Maximum Power Transfer requires $R_L = R_{th}$, giving $P_{max} = V_{th}^2/(4R_{th})$ at 50% efficiency.

CHAPTER 2

Fundamentals of A.C. Circuits

2.1 Introduction and Theoretical Scope

Alternating Current (AC) waveforms represent the standard medium for global energy generation, high-voltage transmission, industrial distribution, and electronic signal processing. The inherent advantage of sinusoidal alternating signals lies in their mathematical preservation under linear differentiation and integration, as well as the ease of voltage transformation via electromagnetic induction. This chapter develops sinusoidal time-domain descriptors, phasor algebra, complex impedance, resonance in series RLC circuits, power factor optimization, and balanced polyphase three-phase systems.

2.2 Sinusoidal Waveform Fundamentals

2.2.1 Time-Domain Characterization

A general time-varying sinusoidal voltage is expressed as:

$$v(t) = V_m \sin(\omega t + \theta) \tag{2.1}$$

- **Peak Amplitude (V_m)**: Maximum deviation from the zero-reference axis.
- **Peak-to-Peak Amplitude (V_{p-p})**: $V_{p-p} = 2V_m$.
- **Angular Frequency (ω)**: $\omega = 2\pi f = \frac{2\pi}{T}$ in radians per second (rad/s).
- **Cyclic Frequency (f)**: $f = 1/T$ in Hertz (Hz), where T is the periodic time in seconds.
- **Phase Angle (θ)**: Angular argument at time $t = 0$.

2.2.2 Effective (RMS) and Average Values

Definition: Root Mean Square (RMS) Value

The Root Mean Square (RMS or effective) value of an alternating current represents the equivalent DC current that produces the identical heating effect (I^2R) in an ohmic resistance over one complete period:

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v(t)^2 dt} \tag{2.2}$$

For a pure sinusoidal signal $v(t) = V_m \sin(\omega t)$:

$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} V_m^2 \sin^2(\theta) d\theta} = \sqrt{\frac{V_m^2}{2\pi}\int_0^{2\pi} \frac{1-\cos(2\theta)}{2} d\theta} = \frac{V_m}{\sqrt{2}} \approx 0.7071 V_m \tag{2.3}$$

Definition: Average Value, Form Factor, and Crest Factor

For symmetrical alternating waveforms, the average over a full period is zero. Hence, the **average value** is evaluated over one half-cycle:

$$V_{avg} = \frac{1}{\pi}\int_0^{\pi} V_m \sin(\theta) d\theta = \frac{V_m}{\pi}[-\cos\theta]_0^{\pi} = \frac{2V_m}{\pi} \approx 0.6366 V_m \tag{2.4}$$

The **Form Factor** (k_f) characterizes waveform shape:

$$k_f = \frac{V_{rms}}{V_{avg}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} \approx 1.11 \quad \text{(Pure Sine Wave)} \tag{2.5}$$

The **Peak (Crest) Factor** (k_p) characterizes peak stress:

$$k_p = \frac{V_m}{V_{rms}} = \frac{V_m}{V_m/\sqrt{2}} = \sqrt{2} \approx 1.414 \quad \text{(Pure Sine Wave)} \tag{2.6}$$

2.3 Phasor Representation and Complex Impedance

By Euler's identity, $e^{j\theta} = \cos\theta + j\sin\theta$. A sinusoidal signal $v(t) = V_m\cos(\omega t + \theta)$ can be viewed as the real projection of a counter-clockwise rotating vector:

$$v(t) = \text{Re}\left\{V_m e^{j(\omega t+\theta)}\right\} = \text{Re}\left\{\sqrt{2}\mathbf{V}e^{j\omega t}\right\} \tag{2.7}$$

where $\mathbf{V} = V_{rms}\angle\theta = V_{rms}(\cos\theta + j\sin\theta)$ is the complex **phasor**. Differentiation in the time domain corresponds to multiplication by $j\omega$ in the frequency domain: $\frac{d}{dt} \longleftrightarrow j\omega$.

Definition: Complex Impedance and Admittance

Impedance $\mathbf{Z} = R + jX$ represents the frequency-domain generalization of resistance:

$$\text{Pure Resistance: } \mathbf{Z}_R = R\angle 0^\circ = R + j0 \tag{2.8}$$

$$\text{Pure Inductance: } \mathbf{Z}_L = j\omega L = \omega L\angle 90^\circ \implies X_L = \omega L = 2\pi f L \quad [\Omega] \tag{2.9}$$

$$\text{Pure Capacitance: } \mathbf{Z}_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = \frac{1}{\omega C}\angle -90^\circ \implies X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \quad [\Omega] \tag{2.10}$$

In a series RLC circuit:

$$\mathbf{Z} = R + j(X_L - X_C) = |\mathbf{Z}|\angle\phi, \quad |\mathbf{Z}| = \sqrt{R^2 + (X_L - X_C)^2}, \quad \phi = \arctan\left(\frac{X_L - X_C}{R}\right) \tag{2.11}$$

2.4 Resonance in Series RLC Circuits

Definition: Resonant State

Resonance occurs in an AC circuit containing reactive elements when the net terminal reactance is identically zero ($X_L = X_C$). Under this condition, the input voltage and current are in phase ($\phi = 0$), the power factor is unity ($\cos\phi = 1.0$), and the circuit impedance attains its absolute minimum: $\mathbf{Z}_0 = R$.

Theorem / Law: Resonance Parameters: Frequency, Quality Factor, Bandwidth

1. **Resonant Frequency**:

$$\omega_0 L = \frac{1}{\omega_0 C} \implies \omega_0 = \frac{1}{\sqrt{LC}} \text{ [rad/s]}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ [Hz]} \tag{2.12}$$

2. **Quality Factor (Q_0)**: Ratio of reactive energy stored to energy dissipated per cycle:

$$Q_0 = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{V_L(\omega_0)}{V} = \frac{V_C(\omega_0)}{V} \tag{2.13}$$

   Since $Q_0 > 1$ in typical communication and tuning circuits, series resonance produces voltage magnification across L and C ($V_L = V_C = Q_0 V_{source}$).

3. **Bandwidth (BW)**: Frequency span between half-power (-3 dB) points where current drops to $I_0/\sqrt{2}$:

$$BW = \Delta f = f_2 - f_1 = \frac{f_0}{Q_0} = \frac{R}{2\pi L} \text{ [Hz]}, \qquad \Delta\omega = \frac{R}{L} \text{ [rad/s]} \tag{2.14}$$

4. **Half-Power Frequencies**:

$$\omega_1, \omega_2 = \mp\frac{R}{2L} + \sqrt{\omega_0^2 + \left(\frac{R}{2L}\right)^2} \tag{2.15}$$

   For high-Q circuits ($Q_0 \geq 10$): $f_1 \approx f_0 - \frac{BW}{2}$ and $f_2 \approx f_0 + \frac{BW}{2}$.

2.5 Power in AC Circuits and Power Factor Correction

2.5.1 The Complex Power Framework

In AC networks, instantaneous power pulsates at twice the supply frequency (2ω). Decomposing this response yields active, reactive, and apparent components:

$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = P + jQ = S\angle\phi \tag{2.16}$$

- **Active / Real Power (P)**: The average rate of energy transferred and converted into useful mechanical, thermal, or optical work:

$$P = V_{rms} I_{rms} \cos\phi \quad \text{[Watts, W]} \tag{2.17}$$

- **Reactive Power (Q)**: The peak rate of energy oscillating between electromagnetic fields and the generator without doing net work:

$$Q = V_{rms} I_{rms} \sin\phi \quad \text{[Volt-Amperes Reactive, VAR]} \tag{2.18}$$

- **Apparent Power (S)**: The total product of RMS voltage and current:

$$S = V_{rms} I_{rms} = \sqrt{P^2 + Q^2} \quad \text{[Volt-Amperes, VA]} \tag{2.19}$$

- **Power Factor (pf)**: The ratio of real power to apparent power:

$$\cos\phi = \frac{P}{S} = \frac{R}{|\mathbf{Z}|} \tag{2.20}$$

  Lagging for inductive circuits (current lags voltage); leading for capacitive circuits (current leads voltage).

2.5.2 Power Factor Correction

Low lagging power factors draw excessive line current ($I = P/(V\cos\phi)$), inducing higher I^2R transmission line losses and voltage drop. To correct the power factor from $\cos\phi_1$ to $\cos\phi_2$ without altering active power P, a shunt capacitor bank is installed:

$$Q_C = P(\tan\phi_1 - \tan\phi_2) \implies C = \frac{P(\tan\phi_1 - \tan\phi_2)}{\omega V_{rms}^2} \tag{2.21}$$

2.6 Three-Phase Balanced Systems

2.6.1 Star (Wye, Y) and Delta ($\triangle$) Topologies

Three-phase generators produce three equal sinusoidal voltages displaced mutually by 120° ($2\pi/3$ rad):

$$v_A(t) = \sqrt{2}V_{ph}\sin(\omega t), \quad v_B(t) = \sqrt{2}V_{ph}\sin(\omega t - 120°), \quad v_C(t) = \sqrt{2}V_{ph}\sin(\omega t - 240°) \tag{2.22}$$

Theorem / Law: Star (Y) Relations

In a balanced three-phase star system with phase voltages V_{AN}, V_{BN}, V_{CN}:

$$\mathbf{V}_{AB} = \mathbf{V}_{AN} - \mathbf{V}_{BN} = V_{ph}\angle 0° - V_{ph}\angle -120° = \sqrt{3}V_{ph}\angle 30° \tag{2.23}$$

Hence:

$$V_L = \sqrt{3}V_{ph}, \qquad I_L = I_{ph} \tag{2.24}$$

The line voltage **leads** the phase voltage by 30°. For balanced loads, the neutral current is zero: $I_N = I_A + I_B + I_C = 0$.

Theorem / Law: Delta ($\triangle$) Relations

In a balanced three-phase delta system:

$$V_L = V_{ph}, \qquad I_L = \sqrt{3}I_{ph} \tag{2.25}$$

The line current **lags** the corresponding phase current by 30°.

2.6.2 Three-Phase Power and the Two-Wattmeter Method

For both star and delta balanced configurations:

$$P_{3\phi} = \sqrt{3}V_L I_L \cos\phi = 3V_{ph} I_{ph} \cos\phi, \qquad Q_{3\phi} = \sqrt{3}V_L I_L \sin\phi, \qquad S_{3\phi} = \sqrt{3}V_L I_L \tag{2.26}$$

By Blondel's Theorem, the active power in an n-wire system requires $n-1$ wattmeters. In a 3-phase, 3-wire system, **two wattmeters** suffice:

$$W_1 = V_L I_L \cos(30^\circ - \phi), \qquad W_2 = V_L I_L \cos(30^\circ + \phi) \tag{2.27}$$

Sum of readings:

$$W_1 + W_2 = V_L I_L[\cos(30^\circ - \phi) + \cos(30^\circ + \phi)] = \sqrt{3}V_L I_L \cos\phi = P_{3\phi} \tag{2.28}$$

Difference of readings:

$$W_1 - W_2 = V_L I_L[\cos(30^\circ - \phi) - \cos(30^\circ + \phi)] = V_L I_L \sin\phi \tag{2.29}$$

Dividing the difference by the sum:

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{\sin\phi}{\sqrt{3}\cos\phi} = \frac{\tan\phi}{\sqrt{3}} \implies \tan\phi = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2} \tag{2.30}$$

2.7 Chapter Summary

- Sine wave measures: $V_{rms} = V_m/\sqrt{2}$, $V_{avg} = 2V_m/\pi$, $k_f = 1.11$, $k_p = 1.414$.
- Impedances: $X_L = \omega L$, $X_C = 1/(\omega C)$, $\mathbf{Z} = R + j(X_L - X_C)$.
- Resonance: $\omega_0 = 1/\sqrt{LC}$, $Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{L/C}$, $BW = f_0/Q = R/(2\pi L)$.
- Power: $P = VI\cos\phi$, $Q = VI\sin\phi$, $S = VI = \sqrt{P^2 + Q^2}$.
- 3-Phase: Star ($V_L = \sqrt{3}V_{ph}, I_L = I_{ph}$); Delta ($V_L = V_{ph}, I_L = \sqrt{3}I_{ph}$).
- Two-wattmeter: $P = W_1 + W_2$, $\tan\phi = \sqrt{3}(W_1 - W_2)/(W_1 + W_2)$.

CHAPTER 3

Fundamentals of Electrical Machines

3.1 Electromagnetic Foundations and Coupling

Electrical machines convert energy between the electrical and mechanical domains, or transform AC electrical power across voltage levels via magnetic coupling. The underlying physics rests on Faraday's law of electromagnetic induction, Lenz's law, Ampere's circuital law, and the Lorentz force principle.

3.1.1 Magnetic Coupling and Dot Convention

When two coils share a mutual magnetic flux:

$$e_1 = -L_1 \frac{di_1}{dt} \mp M \frac{di_2}{dt}, \qquad e_2 = -L_2 \frac{di_2}{dt} \mp M \frac{di_1}{dt} \tag{3.1}$$

where $M = k\sqrt{L_1 L_2}$ is the mutual inductance and $k \in [0, 1]$ is the coupling coefficient.

Definition: Dot Convention

- If currents enter both dotted terminals, the mutually induced voltages **aid** the self-induced voltages ($L_{eq} = L_1 + L_2 + 2M$).
- If current enters one dotted terminal and leaves the other, the mutual coupling **opposes** the self-inductance ($L_{eq} = L_1 + L_2 - 2M$).

3.2 The Single-Phase Transformer

3.2.1 Construction and Principle of Operation

A transformer is a static electromagnetic device transferring electrical power between two or more circuits without a change in frequency. It consists of high-permeability silicon steel laminated cores and insulated copper windings.

Theorem / Law: EMF Equation of a Transformer

Let the core flux vary sinusoidally: $\Phi(t) = \Phi_m \sin(\omega t) = B_m A \sin(2\pi f t)$. By Faraday's Law, the induced EMF in a coil of N turns is:

$$e(t) = -N\frac{d\Phi}{dt} = -N\omega\Phi_m \cos(\omega t) = 2\pi f N \Phi_m \sin(\omega t - 90^\circ) \tag{3.2}$$

The maximum induced EMF is $E_m = 2\pi f N \Phi_m$. The RMS value of induced EMF is:

$$E_{rms} = \frac{E_m}{\sqrt{2}} = \frac{2\pi}{\sqrt{2}} f N \Phi_m = \sqrt{2}\pi f N \Phi_m \approx 4.44 f N \Phi_m \tag{3.3}$$

Hence, for primary and secondary windings:

$$E_1 = 4.44 f N_1 \Phi_m, \qquad E_2 = 4.44 f N_2 \Phi_m \implies \frac{E_2}{E_1} = \frac{N_2}{N_1} = k \tag{3.4}$$

3.2.2 Losses, Efficiency, and Voltage Regulation

- **Core / Iron Losses (P_i)**: Independent of load current, measured via the Open-Circuit (OC) test:

$$P_i = P_h + P_e \tag{3.5}$$

$$P_h = k_h f B_m^{1.6} V \quad \text{(Hysteresis loss, minimized by high-silicon grain-oriented steel)} \tag{3.6}$$

$$P_e = k_e f^2 B_m^2 t^2 V \quad \text{(Eddy current loss, minimized by laminations of thickness } t \approx 0.35 \text{ mm)} \tag{3.7}$$

- **Copper / Ohmic Losses (P_{cu})**: Variable I^2R loss, measured via the Short-Circuit (SC) test:

$$P_{cu}(x) = x^2 P_{cu,FL} \quad \text{where } x = \frac{I_2}{I_{2,FL}} \tag{3.8}$$

- **Transformer Efficiency (η)**:

$$\eta = \frac{\text{Output Power}}{\text{Output Power} + \text{Losses}} = \frac{x S \cos\phi}{x S \cos\phi + P_i + x^2 P_{cu,FL}} \times 100\% \tag{3.9}$$

- **Condition for Maximum Efficiency**: Differentiating η with respect to load current and setting to zero yields:

$$P_i = x^2 P_{cu,FL} \implies x = \sqrt{\frac{P_i}{P_{cu,FL}}} \tag{3.10}$$

  Maximum efficiency occurs when the variable copper loss equals the constant core loss.

- **Voltage Regulation**:

$$\%VR = \frac{V_{2,NL} - V_{2,FL}}{V_{2,FL}} \times 100\% \approx \frac{I_2(R_{e2}\cos\phi \pm X_{e2}\sin\phi)}{V_{2,FL}} \times 100\% \tag{3.11}$$

  ($+$ for lagging power factor, $-$ for leading power factor).

3.2.3 Transformer on D.C. and Auto-Transformers

Why Transformers Fail on D.C.

In a steady DC circuit, $\frac{d\Phi}{dt} = 0$, so no counter-EMF ($E_1 = 0$) is generated. The primary current is limited solely by tiny winding resistance: $I_{dc} = V/R_1$. This draws massive destructive currents ($20 - -50\times$ rated), driving the iron core into deep magnetic saturation and incinerating the primary windings within seconds.

Definition: Auto-Transformer

An auto-transformer utilizes a single continuous tapped winding common to both primary and secondary circuits. The weight of copper in an auto-transformer compared to a conventional two-winding transformer is:

$$\frac{W_{\text{auto}}}{W_{\text{2-wdg}}} = 1 - k \quad \left(k = \frac{V_2}{V_1} < 1\right) \implies \text{Copper Saved} = k \cdot W_{\text{2-wdg}} \tag{3.12}$$

3.3 Direct Current (DC) Machines

3.3.1 Operating Principles and Fleming's Rules

- **Generator Action**: Governed by **Fleming's Right-Hand Rule** (Thumb: Motion, First finger: Field, Second finger: Induced current).
- **Motor Action**: Governed by **Fleming's Left-Hand Rule** (Thumb: Force, First finger: Field, Second finger: Current).
- **Back-EMF Equation**:
$$E_b = \frac{P\Phi ZN}{60A} \tag{3.13}$$
(P: poles, Φ: flux/pole, Z: armature conductors, N: speed in rpm, A: parallel paths, $A = 2$ for wave, $A = P$ for lap).
- **Electromagnetic Torque**:
$$T_e = \frac{1}{2\pi}\left(\frac{P\Phi ZI_a}{A}\right) = 0.159 \cdot \frac{P\Phi ZI_a}{A} \propto \Phi I_a \quad [\text{N} \cdot \text{m}] \tag{3.14}$$
- **Speed Equation**:
$$N = \frac{60A}{PZ}\frac{E_b}{\Phi} \propto \frac{V - I_a R_a}{\Phi} \tag{3.15}$$

3.3.2 Classification and Starting of DC Motors

- **DC Shunt Motor**: Field in parallel with armature. Nearly constant speed from no-load to full-load. Used for machine tools, lathes, centrifugal pumps.
- **DC Series Motor**: Field in series with armature ($I_a = I_{se} \implies T \propto I_a^2$). High starting torque. Never run without load because at no load $\Phi \to 0 \implies N \to \infty$ (catastrophic runaway). Used for cranes, hoists, traction.
- **Motor Starters**: At standstill ($N = 0 \implies E_b = 0$), $I_{start} = V/R_a \approx 25 \times I_{FL}$. A **3-point** or **4-point starter** inserts series resistors during acceleration to protect commutator brushes.

3.4 Three-Phase and Single-Phase AC Induction Motors

3.4.1 Rotating Magnetic Field (RMF) and Operating Principle

When balanced three-phase currents flow through windings displaced by 120° in space, they establish a rotating magnetic field of constant magnitude $B_R = \frac{3}{2}B_m$ rotating at synchronous speed:

$$N_s = \frac{120f}{P} \quad [\text{rpm}] \tag{3.16}$$

This field sweeps across rotor conductors, inducing currents that generate rotor torque in the direction of the field (Lenz's Law).

3.4.2 Slip and Torque–Speed Characteristics

The relative motion is quantified by the **slip** (s):

$$s = \frac{N_s - N}{N_s} \iff N = N_s(1 - s) \tag{3.17}$$

At standstill ($N = 0$): $s = 1.0$. At synchronous speed ($N = N_s$): $s = 0$. The rotor frequency is $f_r = sf$. The electromagnetic torque is:

$$T \propto \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2} \tag{3.18}$$

Maximum torque (T_{max}) occurs at slip $s_{mT} = R_2/X_2$:

$$T_{max} \propto \frac{E_2^2}{2X_2} \tag{3.19}$$

T_{max} is independent of rotor resistance R_2, but increasing R_2 shifts the peak torque toward starting ($s = 1$).

3.4.3 Single-Phase Induction Motors

A single-phase pulsating magnetic field can be resolved by **Double Revolving Field Theory** into two counter-rotating fields of equal magnitude $\frac{1}{2}\Phi_m$. At standstill, forward and backward torques are identical ($T_f = T_b$), yielding zero starting torque ($T_{start} = 0$). To initiate rotation, an auxiliary winding with a series capacitor is added to create a temporary two-phase rotating field (capacitor-start, split-phase, or shaded-pole motors).

3.5 Chapter Summary

- Transformer EMF: $E = 4.44fN\Phi_m$, turns ratio $k = V_2/V_1 = I_1/I_2$.
- Max efficiency condition: $P_i = x^2 P_{cu,FL}$.
- DC motors: $E_b = P\Phi ZN/(60A)$, $T \propto \Phi I_a$, $N \propto E_b/\Phi$.
- Induction motors: Synchronous speed $N_s = 120f/P$, slip $s = (N_s - N)/N_s$, $f_r = sf$.

CHAPTER 4

Fundamentals of Semiconductor Devices

4.1 Introduction and Energy Band Principles

Semiconductor physics enables the active control of charge carriers that defines modern solid-state electronics. Conduction in semiconductors depends on both negative electrons and positive holes. By selectively doping pure intrinsic tetravalent crystals (Silicon, Germanium) with pentavalent donors (P, As) or trivalent acceptors (B, In), n-type and p-type materials are formed.

4.2 The PN Junction Diode

4.2.1 Equilibrium and Bias Conditions

When p-type and n-type semiconductors are metallurgically joined, diffusion of electrons into the p-region and holes into the n-region creates a **depletion region** stripped of mobile carriers, leaving exposed uncompensated donor (N_D^+) and acceptor (N_A^-) ions. This establishes a built-in contact potential:

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right), \qquad V_T = \frac{k_B T}{q} \approx 26 \text{ mV at } 300 \text{ K} \tag{4.1}$$

- **Forward Bias** ($V_D > 0$): External electric field opposes the built-in barrier, shrinking the depletion width and permitting massive majority carrier diffusion:

$$I_D = I_s\left(e^{V_D/(\eta V_T)} - 1\right) \quad \text{(Shockley Equation)} \tag{4.2}$$

  where I_s is the reverse saturation current ($\approx$ pA for Si, $\approx \mu$A for Ge), and η is the ideality factor. Cut-in (knee) voltages: ≈ 0.7 V for Si, ≈ 0.3 V for Ge.
- **Reverse Bias** ($V_D < 0$): External field widens the depletion width, halting majority diffusion; only a tiny thermally generated minority leakage current I_s flows ($I_D \approx -I_s$).

4.2.2 Reverse Breakdown: Zener vs Avalanche

- **Zener Breakdown ($V_Z < 5$ V)**: Occurs in heavily doped junctions with narrow depletion regions (< 10 nm). Intense electric fields ($> 10^6$ V/m) cause quantum mechanical tunneling of valence electrons into the conduction band. Exhibits a **negative temperature coefficient**.
- **Avalanche Breakdown ($V_Z > 6$ V)**: Occurs in lightly doped junctions with wide depletion regions. Accelerated minority carriers gain kinetic energy to impact-ionize covalent lattice bonds, generating an exponential carrier cascade. Exhibits a **positive temperature coefficient**.

4.3 Specialized Diodes and Testing Procedures

4.3.1 Zener Diode as a Voltage Regulator

Operating in the reverse breakdown regime, a Zener diode maintains an almost constant voltage V_Z over wide variations of input voltage and load current. The design of the series current-limiting resistor R_S satisfies:

$$R_S = \frac{V_{in,min} - V_Z}{I_{L,max} + I_{Z,min}} \tag{4.3}$$

where $I_{Z,min}$ is the minimum knee current ensuring stable operation.

4.3.2 Varactor Diode (Varicap)

Under reverse bias, the depletion layer acts as a dielectric between the conductive p and n regions, forming a voltage-variable transition capacitance:

$$C_T(V_R) = \frac{\varepsilon_s A}{W} = \frac{C_0}{(1 + V_R/V_0)^m} \quad (m \approx 0.5 \text{ for abrupt junctions}) \tag{4.4}$$

Widely deployed in voltage-controlled oscillators (VCOs), phase-locked loops (PLLs), and RF frequency synthesis.

4.3.3 Digital Multimeter (DMM) Testing of Diodes and BJTs

Using the diode-check function of a DMM:

- **Healthy Diode**: Forward bias displays the barrier potential ($0.6 - -0.7$ V for Si, $0.2 - -0.3$ V for Ge). Reverse bias displays over-range ('OL' or '1').
- **Faulty Diode**: Reading near zero in both directions indicates an internal short; reading 'OL' in both directions indicates an open circuit.
- **BJT Testing**: Modeled as two back-to-back PN junctions (Base-Emitter and Base-Collector). For an NPN transistor, placing the red (+) probe on the base should show $0.6 - -0.7$ V drop to both emitter and collector.

4.4 Rectifier Circuits and Power Conversion

4.4.1 Topological Comparison of Rectifiers

Metric	Half-Wave	Center-Tap Full-Wave	Bridge Rectifier
Number of Diodes	1	2	4
DC Output Voltage V_{dc}	$\frac{V_m}{\pi} \approx 0.318V_m$	$\frac{2V_m}{\pi} \approx 0.637V_m$	$\frac{2V_m}{\pi} \approx 0.637V_m$
Ripple Factor γ	1.21	0.482	0.482
Ripple Frequency	f	$2f$	$2f$
Max Efficiency η_{max}	40.6%	81.2%	81.2%
Peak Inverse Voltage (PIV)	V_m	$2V_m$	V_m
Transformer Utilization	0.287	0.672	0.812

With a shunt capacitor filter C, the ripple factor drops significantly:

$$\gamma = \frac{1}{4\sqrt{3}fCR_L} \quad \text{(Full-wave)}, \qquad V_{dc} = V_m - \frac{I_{dc}}{4fC} \tag{4.5}$$

4.5 The MOSFET: Metal-Oxide-Semiconductor Field-Effect Transistor

4.5.1 Structure and Operational Regimes (N-Channel Enhancement)

The MOSFET is a unipolar, voltage-controlled device where an electric field applied to an insulated polysilicon gate modulates the conductivity of an inverted electron channel between source and drain.

- **Cutoff Regime** ($V_{GS} < V_{th}$): No inversion channel forms; $I_D = 0$.
- **Triode / Ohmic Regime** ($V_{GS} \geq V_{th}$ and $V_{DS} < V_{GS} - V_{th}$): Channel connects source to drain continuously:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{1}{2} V_{DS}^2 \right] \tag{4.6}$$

- **Saturation Regime** ($V_{GS} \geq V_{th}$ and $V_{DS} \geq V_{GS} - V_{th}$): The channel pinches off at the drain end, and current saturates independently of V_{DS}:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \tag{4.7}$$

  where λ represents channel-length modulation.

- **Transconductance**:

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS}} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th}) = \frac{2 I_D}{V_{GS} - V_{th}} \tag{4.8}$$

4.6 Chapter Summary

- Diode current: $I_D = I_s(e^{V_D/(\eta V_T)} - 1)$, $V_T \approx 26$ mV.
- Zener breakdown (< 5 V, tunneling, negative temp coeff); Avalanche (> 6 V, impact ionization, positive temp coeff).
- Full-wave bridge rectifier: $\eta_{max} = 81.2\%$, $\gamma = 0.482$, PIV $= V_m$.
- MOSFET saturation current: $I_D = \frac{1}{2} k_n' (W/L)(V_{GS} - V_{th})^2$.

CHAPTER 5

Fundamentals of Filters and Operational Amplifier

5.1 Frequency-Selective Filters

Filters are frequency-selective circuits that allow signals within a specified transmission passband to pass with minimal attenuation, while suppressing frequencies in the stopband.

5.1.1 First-Order Passive RC Filters

- **Low-Pass RC Filter**: Formed by series resistor R and shunt capacitor C:

$$H(j\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j(f/f_c)} \tag{5.1}$$

  where the cutoff (-3 dB) frequency is $f_c = \frac{1}{2\pi RC}$. At $f = f_c$, magnitude $|H| = 1/\sqrt{2} = 0.707$ (-3 dB) and phase $\theta = -45°$. For frequencies $f \gg f_c$, attenuation rolls off at -20 dB/decade (-6 dB/octave).

- **High-Pass RC Filter**: Formed by series capacitor C and shunt resistor R:

$$H(j\omega) = \frac{j\omega RC}{1 + j\omega RC} = \frac{j(f/f_c)}{1 + j(f/f_c)} \implies f_c = \frac{1}{2\pi RC} \tag{5.2}$$

  At $f = f_c$, magnitude $|H| = 0.707$ and phase $\theta = +45°$.

- **Band-Pass and Band-Stop Filters**: Cascaded combinations of high-pass and low-pass stages define passbands or notch bands with bandwidth $BW = f_{c2} - f_{c1}$ and center frequency $f_0 = \sqrt{f_{c1} f_{c2}}$.

5.2 The Operational Amplifier (Op-Amp)

5.2.1 Ideal Characteristics and Virtual Ground

The operational amplifier is a direct-coupled, high-gain differential voltage amplifier.

Definition: Ideal Op-Amp Parameters

- Differential Open-Loop Gain: $A_{OL} \to \infty$.
- Input Resistance: $R_{in} \to \infty \implies$ Input bias currents $I^+ = I^- = 0$.
- Output Resistance: $R_{out} = 0$.
- Bandwidth: $BW \to \infty$.
- Slew Rate: $SR \to \infty$.

- Common-Mode Rejection Ratio: CMRR $\to \infty$.
- Input Offset Voltage: $V_{os} = 0$.

Theorem / Law: Virtual Ground / Virtual Short Principle

In an op-amp circuit operating with stable negative feedback and unsaturated output:

$$v^+ - v^- = \frac{v_o}{A_{OL}} \xrightarrow{A_{OL}\to\infty} 0 \implies v^- \approx v^+ \tag{5.3}$$

If the non-inverting terminal (v^+) is grounded (0 V), the inverting terminal (v^-) is held at **virtual ground** potential without any physical copper connection to ground.

5.2.2 Fundamental Linear Op-Amp Configurations

- **Inverting Amplifier**: Applying KCL at the virtual ground node ($v^- = 0$):

$$\frac{v_{in} - 0}{R_1} + \frac{v_o - 0}{R_f} = 0 \implies v_o = -\frac{R_f}{R_1} v_{in}, \qquad A_v = -\frac{R_f}{R_1} \tag{5.4}$$

  The input impedance seen by the source is strictly $R_{in} = R_1$.

- **Non-Inverting Amplifier**: Applying voltage division at the inverting node ($v^- = v^+ = v_{in}$):

$$v_{in} = v_o \frac{R_1}{R_1 + R_f} \implies v_o = \left(1 + \frac{R_f}{R_1}\right) v_{in}, \qquad A_v = 1 + \frac{R_f}{R_1} \tag{5.5}$$

  The input impedance is exceptionally high ($R_{in} \to \infty$).

- **Voltage Follower / Unity-Gain Buffer**: Setting $R_f = 0$ and $R_1 = \infty$ yields $v_o = v_{in}$ ($A_v = 1$). Acts as an impedance matching buffer between high-impedance sensors and low-impedance loads.

- **Inverting Summing Amplifier**:

$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \cdots + \frac{R_f}{R_n} v_n\right) \tag{5.6}$$

- **Difference Amplifier (Subtractor)**:

$$v_o = \frac{R_2}{R_1}(v_2 - v_1) \quad \text{when } \frac{R_4}{R_3} = \frac{R_2}{R_1} \tag{5.7}$$

5.2.3 Dynamic Op-Amp RC Circuits

- **Ideal Integrator**: Capacitor in the feedback branch ($i_C = C\frac{dv_o}{dt}$):

$$\frac{v_{in}}{R} + C\frac{dv_o}{dt} = 0 \implies v_o(t) = -\frac{1}{RC}\int_0^t v_{in}(\tau)d\tau + v_o(0) \tag{5.8}$$

  In frequency domain: $H(s) = -\frac{1}{sRC}$. Practical integrators place a large shunt resistor R_f across C to stabilize DC gain.

- **Ideal Differentiator**: Capacitor in the input branch ($i_C = C\frac{dv_{in}}{dt}$):

$$C\frac{dv_{in}}{dt} + \frac{v_o}{R} = 0 \implies v_o(t) = -RC\frac{dv_{in}(t)}{dt} \tag{5.9}$$

  In frequency domain: $H(s) = -sRC$. Practical differentiators place a small series resistor R_s with C to suppress high-frequency noise amplification.

5.3 Active Filters and Automotive Applications

5.3.1 Active Low-Pass and High-Pass Filters

Active filters combine op-amps with resistors and capacitors, eliminating the need for bulky inductors and offering input-output buffer isolation with controllable passband gain. For an active first-order low-pass filter:

$$f_c = \frac{1}{2\pi RC}, \qquad A_{v0} = 1 + \frac{R_f}{R_1} \tag{5.10}$$

5.3.2 Op-Amp Comparator and ABS Brake Modulation

In open-loop configuration (no feedback), the op-amp acts as an analog voltage comparator:

$$v_o = \begin{cases} +V_{sat} & \text{if } v^+ > v^- \\ -V_{sat} & \text{if } v^+ < v^- \end{cases} \tag{5.11}$$

Anti-lock Braking System (ABS) Application

In an automobile equipped with ABS, magnetic reluctance wheel sensors produce an alternating voltage proportional to the angular rotation of each wheel. An op-amp comparator compares this rectified sensor voltage with an electronic reference threshold V_{ref} derived from chassis deceleration. When severe braking causes an impending wheel lockup ($v_{wheel} < V_{ref}$), the comparator immediately flips its output to HIGH ($+V_{sat}$), signaling the ECU hydraulic modulator to release brake fluid pressure in high-frequency pulses ($15 - -20$ Hz), preventing skidding and preserving steering control.

5.4 Chapter Summary

- Passive RC cutoff: $f_c = 1/(2\pi RC)$.
- Ideal op-amp: $A_{OL} = \infty$, $R_{in} = \infty$, $R_{out} = 0$; Virtual ground: $v^- \approx v^+$.
- Inverting amplifier: $A_v = -R_f/R_1$; Non-inverting amplifier: $A_v = 1 + R_f/R_1$.
- Integrator: $v_o = -\frac{1}{RC}\int v_{in}dt$; Differentiator: $v_o = -RC\frac{dv_{in}}{dt}$.
- Comparator serves as high-speed threshold detector in safety ABS architectures.

CHAPTER 6

Embedded Systems and IoT Applications

6.1 Embedded System Fundamentals

An embedded system is a dedicated, computer-based electromechanical system designed to perform a specialized set of real-time monitoring and control functions, typically constrained by strict memory, power, latency, and environmental requirements.

6.1.1 Architectural Building Blocks

An embedded system integrates:

1. **Core Processing Unit**: Microcontroller, DSP, or FPGA executing firmware.
2. **Memory Subsystem**: Non-volatile Flash/ROM for program code and volatile SRAM for runtime stack and heap.
3. **Peripherals & I/O**: Timers, Counters, Watchdog Timers, PWM channels, and ADCs.
4. **Communication Buses**: Inter-Integrated Circuit (I^2C), Serial Peripheral Interface (SPI), UART, and Controller Area Network (CAN).
5. **Power Management**: Low-power sleep modes and voltage regulators.

6.2 Microprocessors versus Microcontrollers

Metric	Microprocessor (μP)	Microcontroller (μC)
Integration	CPU cores and caches only	CPU + Flash + RAM + ADC + Timers on a single die
Target Domain	High-throughput, general-purpose computation	Dedicated, real-time control applications
Bus Architecture	External address/data buses (Von Neumann)	Internal high-speed buses (Harvard architecture)
Power Consumption	High (tens to hundreds of Watts)	Ultra-low (milliwatts to microwatts)
Execution Latency	Non-deterministic (OS context switching)	Hard real-time deterministic cycle execution
Cost	High per processor system	Extremely economical (sub-dollar units)
Examples	Intel Core i7, AMD Ryzen, Apple M3	ATmega328P, STM32, ESP32, PIC16F

6.3 The Arduino Ecosystem and Sensor Interfacing

6.3.1 ATmega328P Microcontroller Architecture

The Arduino Uno development board is based on the Microchip ATmega328P 8-bit AVR RISC microcontroller:

- Clock speed: 16 MHz crystal oscillator (62.5 ns cycle instruction time).
- Memory: 32 KB In-System Flash (0.5 KB used by bootloader), 2 KB SRAM, 1 KB EEPROM.
- I/O capabilities: 14 digital pins (pins 0–13, with 6 providing 8-bit hardware PWM at ≈ 490 Hz or 980 Hz), 6 analog input channels (A0–A5) multiplexed to an internal 10-bit Successive Approximation ADC.

6.3.2 Analog-to-Digital Conversion (ADC) Mathematics

The internal ADC maps an analog voltage $V_{in} \in [0, V_{ref}]$ into a discrete 10-bit integer $D \in [0, 1023]$:

$$D = \left\lfloor \frac{V_{in}}{V_{ref}} \times 1024 \right\rfloor \iff V_{in} = \frac{D \times V_{ref}}{1024} \tag{6.1}$$

The quantization resolution (Least Significant Bit voltage, V_{LSB}) is:

$$V_{LSB} = \frac{V_{ref}}{2^n} = \frac{5.0 \text{ V}}{1024} \approx 4.8828 \text{ mV} \tag{6.2}$$

Switching to the internal 1.1 V bandgap reference improves voltage resolution to:

$$V_{LSB,internal} = \frac{1.1 \text{ V}}{1024} \approx 1.0742 \text{ mV} \tag{6.3}$$

6.3.3 Sensor Interfacing Transfer Functions

- **LM35 Linear Analog Temperature Sensor**: Produces a linear voltage output directly proportional to Celsius temperature:

$$V_{out} = 10 \text{ mV}/^\circ\text{C} \times T \implies T(^\circ\text{C}) = \frac{V_{out} \text{ [mV]}}{10} = \frac{D \times (5000/1024)}{10} \tag{6.4}$$

- **HC-SR04 Ultrasonic Range Finder**: Emits a 40 kHz ultrasonic burst and measures echo return time Δt. Using speed of sound $v = 340$ m/s $= 0.034$ cm/μs:

$$\text{Distance } d = \frac{v \times \Delta t_{\text{echo}}}{2} = \frac{\Delta t_{\text{echo}} \text{ [}\mu\text{s]}}{58.2} \quad \text{[cm]} \tag{6.5}$$

6.4 Internet of Things (IoT) and Cloud Applications

6.4.1 Layered IoT Architecture

Modern Internet of Things ecosystems organize edge sensing and cloud analytics into four functional tiers:

1. **Perception Tier**: Transducers, smart sensors, and microcontrollers sample physical metrics.
2. **Edge Tier**: Edge devices filter raw data, perform moving averages, and format payloads into JSON.
3. **Network / Transport Tier**: Telemetry is transmitted over Wi-Fi, Cellular (NB-IoT), or LoRaWAN using lightweight protocols such as **MQTT** (Message Queuing Telemetry Transport) over TLS/TCP.
4. **Cloud Application Tier**: Cloud IoT engines (AWS IoT Core, Azure IoT Hub, Google Cloud IoT) store time-series records and feed analytical dashboards.

6.4.2 Industrial and Societal Application Domains

- **Condition Monitoring in Process Plants**: Continuous vibration, thermal, and acoustic tracking on motor bearings to predict mechanical failure before catastrophic shutdown (Predictive Maintenance / Industry 4.0).
- **Healthcare Telemetry**: Wearable patient telemetry monitoring heart rate (ECG), SpO2, and blood pressure with cloud alert triggers.
- **Precision Agriculture**: Soil moisture probes and automated solenoid valve control optimizing irrigation and fertilizer delivery.
- **Smart Electrical Grids**: Automated Metering Infrastructure (AMI) and synchronized phasor measurement units (PMUs) balancing supply and demand dynamically.

6.5 Chapter Summary

- Microcontroller integrates CPU, Flash, SRAM, and I/O peripherals onto a single silicon chip.
- ATmega328P 10-bit ADC has step size $V_{LSB} = V_{ref}/1024 \approx 4.88$ mV at 5 V.
- LM35 sensor transfer function: 10 mV/°C; Ultrasonic sensor: $d = \Delta t/58.2$ cm.
- IoT architecture: Perception $\rightarrow$ Edge processing $\rightarrow$ MQTT network $\rightarrow$ Cloud analytics.